

RVL-DiffGrid: Robust Vision-Language Diffusion for Counting with Hierarchical Semantic Grids

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Abstract—Crowd counting in highly congested scenes remains challenging, as global semantic cues alone are insufficient for fine-grained density estimation. We propose RVL (Robust Vision-Language)-DiffGrid, a diffusion-enhanced vision-language framework with a dual-branch architecture. The CLIP-based global counting head provides training-time semantic supervision and is discarded at inference, while CLIP encoders are retained only to condition U-Net cross-attention. The diffusion-guided spatial branch encodes the image into latent space, applies a single diffusion step, and extracts multi-scale cross-attention from a frozen text-conditioned U-Net, organized into Hierarchical Semantic Grids (HSG) with structural regularization to stabilize density estimation under scale variation and occlusion. The final count is obtained by spatially integrating the predicted density map. Experiments on five benchmarks demonstrate competitive performance: 51.39/94.36 MAE/RMSE on ShanghaiTech Part A, 5.79/9.12 on Part B, 217.88/268.11 on UCF-CC-50, 75.19/121.78 on UCF-QNRF, and 56.21/215.30 on JHU-Crowd++.

Index Terms—Diffusion Models, Vision-Language Learning, Crowd Counting, Hierarchical Grid Reasoning, Density Map Regression

I. INTRODUCTION

Crowd counting in highly congested scenes remains challenging due to severe occlusion, extreme density variation, and complex backgrounds. CNN-based methods such as MCNN [1], CSRNet [2], and RepSFNet [3] improve robustness through multi-scale and contextual modeling, but their performance degrades in highly dense scenarios. Vision-language models like CLIP [4] provide global semantic priors that enhance scene understanding, yet they lack the spatial precision required for accurate density localization. Meanwhile, diffusion models reveal that cross-attention layers encode people-related spatial patterns, but these signals are often noisy and inconsistent. To address these limitations, we propose *RVL-DiffGrid*, a diffusion-enhanced vision-language framework with two complementary branches. The CLIP-based global counting head provides training-time semantic super-

vision to anchor scene-level understanding and is discarded at inference; CLIP encoders are retained only to condition U-Net cross-attention. The diffusion-guided spatial branch extracts multi-scale cross-attention from a frozen text-conditioned U-Net and organizes them into *Hierarchical Semantic Grids* (HSG) for structured density estimation. The final count is obtained by integrating the predicted density map. This design enables RVL-DiffGrid to leverage semantic guidance during training and structured diffusion priors at inference, without fine-tuning or generative sampling. Experiments on multiple benchmarks demonstrate robust and competitive performance in dense crowd scenarios.

II. RELATED WORK

CNN-Based Methods. MCNN [1] and CSRNet [2] leverage multi-scale and dilated convolutions for density regression, and RepSFNet [3] further introduces structural reparameterization for efficient single-fusion estimation, but they lack global context modeling and struggle under extreme density variation and occlusion.

Key Challenges. Crowd counting faces three core challenges: (i) extreme scale variation, (ii) severe occlusion, and (iii) annotation noise from Gaussian-based density generation, demanding models that jointly capture global semantics and fine-grained spatial structure.

Beyond CNN. DM-Count [5] improves noise robustness via optimal transport but lacks spatial priors. DSGC-Net [6] models long-range dependencies through graph reasoning but without semantic grounding. Lightweight transformer designs such as MicroViT [7] and FaceLiVT [8] reduce attention complexity for edge-friendly deployment, but are not tailored to density regression. Neither line of work leverages pretrained foundation models for semantic guidance.

Vision-Language Models. CLIP [4] provides strong semantic priors, and CountGD [9] extends this to open-world counting, yet lack fine-grained spatial precision for density estimation.

Diffusion Models. CrowdDiff [10] incurs high inference cost from iterative sampling; Hui [11] targets open-world generation without explicit density supervision. Both lack spatial precision or efficiency for regression tasks.

Positioning of RVL-DiffGrid. RVL-DiffGrid repurposes frozen diffusion cross-attention as structured spatial priors for density regression without sampling, fine-tuning, or generative objectives. A single forward pass at $t=500$ activates hierarchical spatial responses with only 0.30M trainable parameters and 13.56 ms latency, matching the efficiency profile required by edge-oriented vision systems [12].

III. METHOD

Figure 1 illustrates the RVL-DiffGrid architecture. The global branch (Figure 1, top) produces a fused image-text embedding and a coarse global count used only during training for semantic supervision. The diffusion branch (Figure 1, bottom) encodes the image into latent space, applies a single diffusion step, and extracts multi-scale latent features and cross-attention maps from a frozen text-conditioned U-Net.

A. CLIP-Based Image-Text Encoding and Fusion

The global branch constructs a multimodal semantic representation from image and text inputs. Given an image I , the CLIP ViT-L/14 image encoder extracts a visual embedding

$$F_v = \text{CLIP}_{\text{img}}(I), \quad (1)$$

and a textual prompt is encoded as

$$E_t = \text{CLIP}_{\text{text}}(\text{prompt}). \quad (2)$$

The embeddings are fused via a lightweight learnable mapping

$$F_{\text{fused}} = \phi(F_v, E_t), \quad (3)$$

where $\phi(\cdot)$ denotes a linear projection or lightweight cross-attention module. The fused embedding provides semantic conditioning for the global counting head and serves as the cross-attention context for the diffusion U-Net.

Both CLIP encoders are initialized from OpenCLIP pretrained weights and kept fully frozen; only $\phi(\cdot)$ and the global counting head are optimized. Text embedding refinement thus refers to learning an adaptive projection into the diffusion conditioning space. The original tokenizer and text encoder of the diffusion model are bypassed, and the projected CLIP embedding is directly injected as cross-attention context into the frozen diffusion U-Net. At inference, only the global counting head is discarded; CLIP encoders and $\phi(\cdot)$ remain active solely to supply this context.

B. Global Counting Loss for Semantic Guidance

The global branch processes F_{fused} to predict a coarse global count $\hat{C}_{\text{global}}$, which serves *only* as an auxiliary supervisory signal during training and is fully discarded at inference. The global counting loss is formally defined as:

$$\mathcal{L}_{\text{global}} = \mathcal{L}_{\text{Huber}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}} + \lambda_{\text{ADR}} \mathcal{L}_{\text{ADR}}, \quad (4)$$

where $\mathcal{L}_{\text{Huber}}$ is the Huber regression loss that stabilizes count prediction under large errors, behaving as ℓ_2 for small residuals and as ℓ_1 for large ones, thereby reducing sensitivity to outlier counts in extremely dense scenes.

a) Relative deviation loss.

The relative loss $\mathcal{L}_{\text{rel}}$ normalizes global count errors with respect to crowd scale, preventing the loss from being dominated by scenes with very large absolute counts:

$$\mathcal{L}_{\text{rel}} = \frac{|\hat{C}_{\text{global}} - C^{gt}|}{C^{gt} + \epsilon}, \quad (5)$$

where C^{gt} denotes the ground-truth count and ϵ is a small constant for numerical stability.

b) Adaptive density regularization.

The adaptive density regularization loss $\mathcal{L}_{\text{ADR}}$ enforces consistency between the global count prediction and the spatially integrated density map:

$$\mathcal{L}_{\text{ADR}} = \left| \hat{C}_{\text{global}} - \sum_{x,y} \hat{D}(x,y) \right|. \quad (6)$$

This term propagates gradients to both the global counting head and the density regression branch, encouraging coherent global-local predictions throughout training.

The global branch does not form an ensemble with the density-based prediction; instead, it refines the text embedding through supervised semantic alignment. Both $\mathcal{L}_{\text{rel}}$ and $\mathcal{L}_{\text{ADR}}$ are applied only during training. At inference, the global head is discarded and the final crowd count $\hat{C}$ is obtained solely by spatially integrating the predicted density map as given by Eq. 16.

C. Diffusion-Guided Latent Encoding and Attention Extraction

The diffusion-guided branch encodes the input image into a compact latent space and exploits frozen diffusion cross-attention as a structured spatial prior. This process proceeds in three sequential steps, with *no denoising objective optimized at any stage* and no gradients propagating into the diffusion backbone.

Step 1 — VAE latent encoding. Given an input image I , a pretrained VAE encoder produces a compact latent representation:

$$z_0 = \text{VAE}_{\text{enc}}(I), \quad (7)$$

which preserves the spatial structure of the scene while substantially reducing the dimensionality of subsequent computations. For diffusion-branch processing, the input is resized to 512×512 before VAE encoding, yielding a $64 \times 64 \times 4$ latent tensor compatible with the pretrained Stable Diffusion U-Net; the CLIP global branch independently operates on 336×336 inputs as specified in Section IV-A. At inference, this encoding is performed once, and all subsequent diffusion-guided processing and density regression operate directly in latent space.

Step 2 — Single forward diffusion step. A single forward diffusion step is applied at a fixed intermediate timestep $t=500$ under the standard 1000-step linear noise schedule:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (8)$$

where $\bar{\alpha}_t$ is the cumulative noise coefficient at timestep t . The choice of $t=500$ reflects a deliberate design trade-off: very early timesteps ($t \ll 500$) yield attention maps that

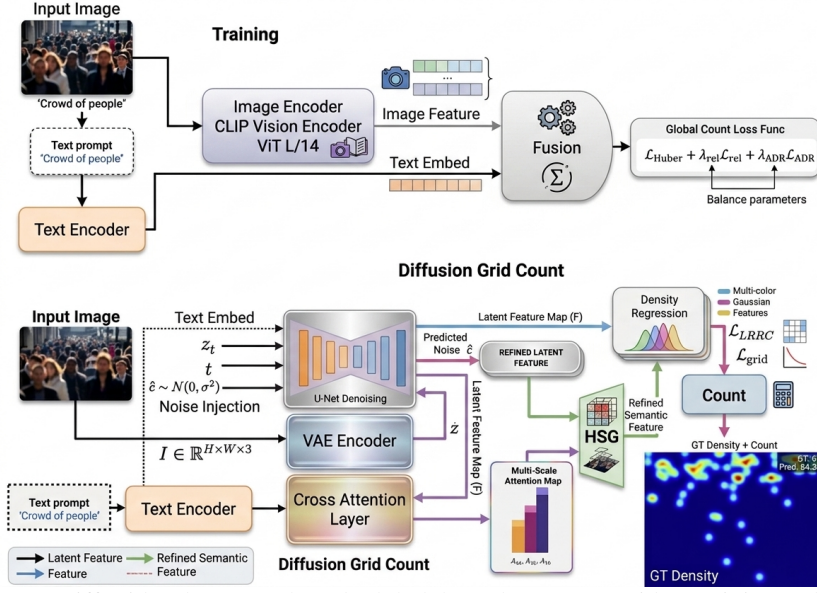


Fig. 1: Overview of RVL-DiffGrid. The CLIP-based global branch (top) provides training-only semantic supervision; the diffusion-guided spatial branch (bottom) extracts multi-scale cross-attention from a frozen U-Net, organized into Hierarchical Semantic Grids (HSG) and fused with refined latent features F_{ref} for density regression.

closely mirror the clean image and lack sufficient semantic abstraction, while very late timesteps ($t \gg 500$) introduce excessive noise that degrades spatial coherence. Empirically, aggregating multiple timesteps yields only marginal gains at higher computational cost; a single step at $t=500$ thus provides the best balance between semantic alignment and efficiency.

Step 3 — Frozen U-Net spatial prior extraction. The noisy latent z_t is passed through a frozen text-conditioned diffusion U-Net, conditioned on the fused image-text embedding E_t :

$$\hat{\epsilon} = \text{UNet}(z_t, \tau(t), E_t), \quad (9)$$

where $\tau(t)$ denotes the timestep embedding and $\hat{\epsilon}$ denotes the U-Net output (predicted noise in the standard diffusion formulation), used here only as a notational anchor: no denoising loss is computed and no gradients propagate into the diffusion backbone. Instead, intermediate multi-scale latent features and cross-attention maps are extracted from the U-Net decoder and treated directly as frozen spatial priors for density regression.

Cross-attention conditioning and frozen diffusion prior

We adopt a pretrained latent diffusion model with its original denoising U-Net kept fully frozen. The fused CLIP embedding is projected through a learnable linear layer to the diffusion context dimension (768) and reshaped into a token sequence of length $N = 77$, matching the default CLIP context length of Stable Diffusion. This sequence serves as the cross-attention context at all U-Net attention layers, bypassing the original diffusion text encoder entirely. These frozen diffusion cues are subsequently organized by the proposed Hierarchical Semantic Grids (Section III-D) for structured density regression.

Latent Feature Extraction and Refinement The frozen diffusion U-Net produces multi-scale latent features at three spatial resolutions:

$$\mathcal{F} = \{F_{64}, F_{32}, F_{16}\}, \quad (10)$$

capturing spatial information from coarse scene layouts to fine-grained local details. At each scale $s \in \{64, 32, 16\}$, cross-attention aligns visual features with the fused image-text semantic cues:

$$A_s = \text{softmax}\left(\frac{Q_s K_s^\top}{\sqrt{d}}\right), \quad s \in \{64, 32, 16\}. \quad (11)$$

The extracted multi-scale features are subsequently passed through a lightweight refinement module $\Phi(\cdot)$ that enhances inter-scale feature coherence and suppresses residual diffusion noise introduced by the single-step forward process:

$$F_{\text{ref}} = \Phi(F_{64}, F_{32}, F_{16}). \quad (12)$$

The refined representation F_{ref} thus consolidates spatial information across all three scales into a stable, noise-suppressed feature map, which is subsequently passed to the HSG module for structured density reasoning.

D. Hierarchical Semantic Grids (HSG) and Density Regression

To mitigate noise and scale inconsistency in diffusion cross-attention maps, we introduce *Hierarchical Semantic Grids (HSG)*, which convert multi-scale diffusion attention into a structured mid-level representation for density regression.

Cross-attention extraction. From the frozen diffusion U-Net, cross-attention maps are extracted at three decoder scales with spatial resolutions $s \in \{64, 32, 16\}$. At each scale, attention maps are averaged across heads and aggregated over all text tokens, yielding stable and prompt-robust spatial responses.

Grid construction and fusion. Each attention map is resized via bilinear interpolation to a fixed grid resolution $g \times g = 16 \times 16$:

$$\hat{G}_s \in \mathbb{R}^{16 \times 16}, \quad s \in \{64, 32, 16\}. \quad (13)$$

Each grid is independently min-max normalized to $[0, 1]$ per image. The normalized grids are fused by a weighted sum:

$$S = \sum_{s \in \{64, 32, 16\}} w_s \cdot \hat{G}_s, \quad (14)$$

where $w_s = \frac{1}{3}$ are fixed uniform weights. The grid resolution is fixed to 16×16 as a trade-off between spatial expressivity and robustness to diffusion noise. Coarser grids tend to lose fine spatial cues, while finer grids are more sensitive to noisy attention responses. Uniform fusion weights are adopted for stability and simplicity; exploring alternative grid sizes and learned fusion weights is left for future work.

Density regression. The resulting grid $S \in \mathbb{R}^{16 \times 16}$ encodes diffusion-derived semantic priors and is concatenated with refined latent features F_{ref} to predict the density map via a lightweight convolutional regressor $\Gamma(\cdot)$:

$$\hat{D} = \Gamma(F_{\text{ref}}, S). \quad (15)$$

The final crowd count is obtained by spatial integration:

$$\hat{C} = \sum_{x,y} \hat{D}(x, y). \quad (16)$$

By jointly leveraging refined diffusion features and structured semantic grids, the regression head promotes spatial coherence while preserving fine-grained localization in highly congested scenes.

E. Training Losses and Final Count Integration

To encourage spatial coherence in the predicted density structure, we introduce a grid smoothness regularization term, denoted **LRRC** (**L**ocal-**R**egional **R**egularity **C**onstraint), implemented as a first-order anisotropic total variation (TV) regularization on the coarse 16×16 hierarchical semantic grid S . It acts as a lightweight structural stabilizer rather than a rigid pixel-level constraint, with no region-level grouping or graph-based computation. Formally, the LRRC loss penalizes abrupt changes between neighboring grid cells:

$$\mathcal{L}_{\text{LRRC}} = \sum_{i=1}^{g-1} \sum_{j=1}^{g-1} (|S(i, j) - S(i+1, j)| + |S(i, j) - S(i, j+1)|), \quad (17)$$

where $S(i, j)$ denotes the semantic grid value at spatial location (i, j) and g denotes the grid resolution.

Grid-level structural alignment. In addition to local smoothness, a grid-level structural alignment loss preserves consistency between predicted density distributions and ground-truth spatial layouts:

$$\mathcal{L}_{\text{grid}} = \sum_k \left| \mathcal{G}_k(\hat{D}) - \mathcal{G}_k(D_{\text{gt}}) \right|, \quad (18)$$

where $\mathcal{G}_k(\cdot)$ computes the integrated density within the k -th grid cell, $\hat{D}$ denotes the predicted density map, and D_{gt} denotes the ground-truth density map. This term enforces coarse regional consistency while avoiding rigid point-level constraints that could over-regularize dense scenes.

Structural regularization. The two structural losses are combined as:

$$\mathcal{L}_{\text{struct}} = \lambda_{\text{LRRC}} \mathcal{L}_{\text{LRRC}} + \lambda_{\text{grid}} \mathcal{L}_{\text{grid}}. \quad (19)$$

Overall objective. The final training objective integrates pixel-wise density regression, global count supervision (Eq. 4), and

grid-level structural regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{dens}} + \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{struct}}, \quad (20)$$

where $\mathcal{L}_{\text{dens}}$ is a standard ℓ_2 density regression loss serving as the primary pixel-wise supervision, $\mathcal{L}_{\text{global}}$ is the auxiliary global count loss in Eq. 4, and $\mathcal{L}_{\text{struct}}$ groups the LRRC and grid alignment regularizers; sub-component weights $(\lambda_{\text{rel}}, \lambda_{\text{ADR}}, \lambda_{\text{LRRC}}, \lambda_{\text{grid}})$ are listed in Table V. At inference, only the density regression branch is retained; global and structural terms act solely as training-time supervision. The final crowd count is obtained by spatially integrating the predicted density map as given by Eq. 16.

IV. EXPERIMENTAL RESULTS

We evaluate RVL-DiffGrid on five benchmarks: ShanghaiTech Part A/B, UCF-CC-50, UCF-QNRF, and JHU-Crowd++, under frozen single-step diffusion and standard density-regression protocols.

A. Training Details and Reproducibility

All experiments use PyTorch on a single NVIDIA RTX 5090 GPU (32GB VRAM). The CLIP encoder (ViT-L/14-336) and diffusion backbone are frozen; only the fusion module and regression heads are trained. CLIP and diffusion branches use 336×336 and 512×512 inputs (cf. Section III-C), with horizontal flipping and random erasing, and no test-time augmentation. Density maps use adaptive Gaussian kernels for ShanghaiTech and UCF, and fixed kernels for JHU-Crowd++. We follow 5-fold cross-validation for UCF-CC-50 and official splits elsewhere; hyperparameters are listed in Table V.

B. Comparison with State-of-the-Art Methods

Table I shows RVL-DiffGrid achieves the best density-regression results on ShanghaiTech Part B (5.79/9.12 MAE/RMSE), UCF-QNRF (75.19/121.78), and the lowest RMSE on UCF-CC-50 (268.11). On Part A, it reaches 51.39 MAE, outperforming the vision-language baseline CLIP-EBC [28] (51.39 vs. 52.5) and remaining close to task-specific methods LIMM (50.8) and DSGC-Net (48.9), which retain the lead on Part A RMSE. On UCF-QNRF, RVL-DiffGrid likewise improves over CLIP-EBC (75.19 vs. 80.3).

C. Ablation Study

On ShanghaiTech Part A (Table II), the Diffusion-Guided Component (DGC) alone reduces MAE from 87.83 to 65.80, and the full model reaches 51.39 MAE.

HSG normalization and fusion weights. Per-image min-max normalization is used by default; global and batch normalization degrade at batch size = 1. Learnable fusion weights give +0.30 MAE and are replaced by uniform weights.

Parameter sensitivity. Table III shows $t=500$ and 16×16 grids yield the best trade-off.

Attention map visualization. Figure 2 shows the standard prompt produces focused activations on crowd-dense regions, while random and removed prompts yield diffuse responses, confirming language as a semantic anchor.

TABLE I: Comparison with SOTA on ShanghaiTech Part A/B, UCF-CC-50, UCF-QNRF, and JHU-Crowd++.

Method	SH Part A		SH Part B		UCF-CC-50		UCF-QNRF		JHU-Crowd++	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
CSRNet [2]	68.2	115.0	10.6	16.0	266.1	397.5	120.3	208.5	—	—
Zhang et al. [13]	181.8	277.7	32.0	49.8	467.0	498.5	—	—	—	—
MCNN [1]	110.2	173.2	26.4	41.3	467.0	498.5	243.5	364.7	—	—
M. Zand et al. [14]	71.4	110.7	9.6	15.0	—	—	—	—	—	—
Multiscale-CNN [15]	83.7	124.5	17.9	32.4	—	—	—	—	—	—
S.D. Khan et al. [16]	77.6	129.7	14.1	21.1	402.3	434.1	218.2	357.4	—	—
PCCNet [17]	73.5	124.0	11.0	19.0	240.0	315.5	149.0	247.0	—	—
SCLNet [18]	67.9	102.9	9.1	14.1	258.9	326.2	109.6	182.5	—	—
Dream [19]	62.6	102.0	7.9	13.4	—	—	94.0	159.8	—	—
ICC [20]	77.97	130.1	8.4	15.2	—	—	—	—	—	—
Wan et al. [21]	61.3	95.4	7.3	11.7	—	—	84.3	147.5	—	—
MSSRG++ [22]	86.6	137.8	—	—	—	—	115.6	213.9	—	—
MPCount [23]	99.6	182.9	11.4	19.7	—	—	165.6	290.4	—	—
PFSNet [24]	—	—	—	—	—	—	—	—	61.2	257.8
UOT [25]	—	—	—	—	—	—	—	—	60.5	252.7
TopoCount [26]	—	—	—	—	—	—	—	—	60.9	267.4
LIMM [27]	50.8	84.2	6.5	10.2	—	—	76.4	125.3	53.0	207.9
DM-Count [5]	59.7	95.7	7.4	11.8	211.0	291.5	85.6	148.3	—	—
DSGC-Net [6]	48.9	77.8	5.9	9.3	—	—	79.3	133.9	—	—
CLIP-EBC [28]	52.5	85.9	6.6	10.5	—	—	80.3	139.3	—	—
RVL-DiffGrid (ours)	51.39	94.36	5.79	9.12	217.88	268.11	75.19	121.78	56.21	215.30

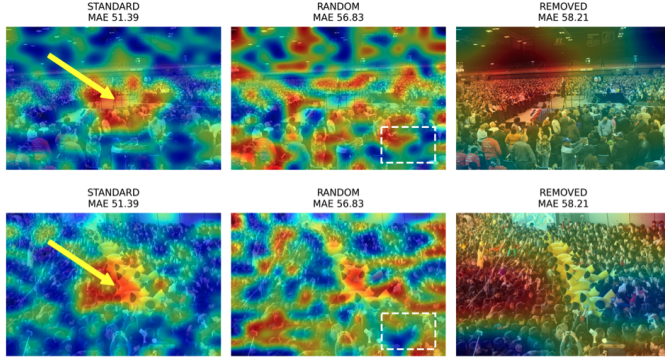


Fig. 2: Cross-attention maps under three prompt conditions on SH Part A; numbers denote full-dataset MAE (cf. Table IV). Standard prompting consistently activates crowd-dense regions; random or removed prompts yield noisier, less structured attention.

Language modality ablation. Table IV shows that removing or randomizing the prompt degrades MAE, while a visual-only token yields intermediate results, confirming that language provides complementary semantic guidance.

TABLE II: Ablation study of RVL-DiffGrid on SHA.

Method	DGC	HSG	LRRC	MAE	RMSE
Baseline (CLIP-only)	—	—	—	87.83	141.01
+ DGC	✓	—	—	65.80	121.70
HSG + LRRC only	—	✓	✓	67.26	100.85
DGC + HSG	✓	✓	—	62.68	97.06
RVL-DiffGrid (full)	✓	✓	✓	51.39	94.36
<i>HSG normalization & fusion weight ablation</i>					
w/ global norm.	✓	✓	✓	54.17	98.83
w/ batch norm.	✓	✓	✓	55.62	100.31
w/ learnable weights	✓	✓	✓	51.69	95.07

Table V summarizes dataset-specific learning rates and loss weights. Congested datasets use larger λ_{LRRC} and λ_{grid} , while ShanghaiTech Part B applies milder regularization. JHU-Crowd++ uses a smaller learning rate and disables both the relative loss ($\lambda_{\text{rel}} = 0$) and grid alignment ($\lambda_{\text{grid}} = 0$), retaining only $\lambda_{\text{LRRC}} = 0.02$ for local smoothness, since heavy

 TABLE III: Parameter sensitivity on SHA. Default: $t=500$, $g=16$, uniform weights.

Parameter	Setting	MAE	RMSE	Parameter	Setting	MAE	RMSE
Timestep t	250	57.43	103.21	Grid g	8×8	54.76	98.53
	500	51.39	94.36		16×16	51.39	94.36
	750	55.87	100.64		32×32	53.21	97.18
Fusion	fixed	51.39	94.36	Fusion	learnable	51.69	95.07

TABLE IV: Prompt ablation on SHA.

Prompt	MAE	RMSE	Prompt	MAE	RMSE
Standard prompt	51.39	94.36	No prompt (removed)	58.21	104.92
("a crowd of people")			Visual-only token	54.67	98.13
Random prompt	56.83	101.47			

occlusion and noisy point annotations make scale-normalized and rigid grid-level supervision unreliable.

TABLE V: Hyperparameter tuning across datasets

Dataset	LR	λ_{ADR}	λ_{rel}	λ_{LRRC}	λ_{grid}
SHA Part A	5×10^{-6}	0.01	0.03	0.05	0.05
SHA Part B	5×10^{-6}	0.01	0.02	0.02	0.03
UCF-CC-50	3×10^{-6}	0.01	0.05	0.10	0.10
UCF-QNRF	1×10^{-5}	0.01	0.04	0.05	0.05
JHU-Crowd++	1×10^{-6}	0.02	0.00	0.02	0.00

D. Efficiency Analysis

Evaluated on a single NVIDIA RTX 5090 GPU (32GB VRAM), batch size = 1; CLIP input 336×336 , diffusion input 512×512 . At inference, the global counting head is discarded; CLIP and VAE encoders remain active to supply cross-attention context and the latent input to the frozen U-Net. With 428.24M total and only 0.30M trainable parameters (123.21 GFLOPs), the diffusion-side pipeline runs in 13.56 ms per image (~ 73.7 FPS): U-Net 10.2 ms ($\sim 75\%$), HSG 0.7 ms, regression head 2.6 ms; one-time CLIP and VAE encoding are not included in this breakdown, confirming real-time practicality with minimal trainable overhead.

E. Qualitative Visualization Across Datasets

Figure 3 shows that RVL-DiffGrid produces smooth, well-localized density maps under extreme crowd densities and

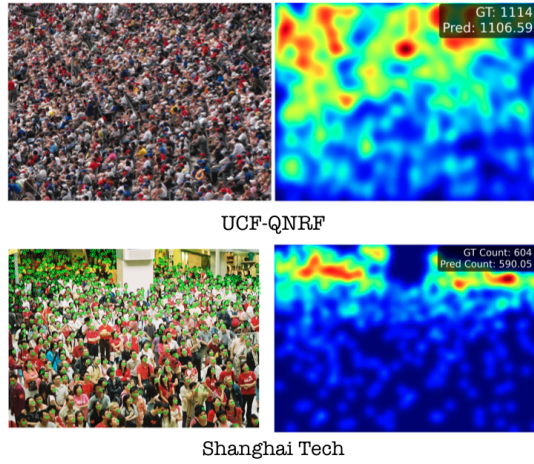


Fig. 3: Qualitative predictions of RVL-DiffGrid on UCF-QNRF and ShanghaiTech Part A, showing spatially coherent density estimation across diverse crowd densities.

large scale variations, confirming strong generalization across diverse density regimes.

V. CONCLUSION

RVL-DiffGrid integrates frozen vision–language semantics with diffusion-guided spatial reasoning by converting cross-attention priors into structured Hierarchical Semantic Grids for density regression. Experiments on five benchmarks demonstrate robust performance across sparse and highly congested scenes without specialized likelihoods or graph-based designs. Ablation studies confirm that diffusion priors, hierarchical grid structuring, and local smoothness regularization contribute complementarily, validating the practicality of leveraging frozen generative and vision–language priors for crowd counting.

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